

Code reference: what each reported number is, and where it is computed

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This maps every quantity in the paper to the script and function that produces it, states the exact definition used, and flags the places where the implementation and the text do not yet correspond. Section 5 is the list I would most like to go through with you.

1 Layout

One module, `lfm.clfm.cdfm_experiment.py`, holds the models, data, evaluator and sampler. Every measurement script imports it with the environment variable `ABLATION_IMPORT=1`, which makes the module raise a sentinel exception after the models, dataset and evaluator are constructed but before it runs its own evaluation. The importing script catches the sentinel and reuses the loaded objects. Consequence worth knowing: only definitions appearing *before* the sentinel are available to measurement scripts.

Script	Produces
<code>curvature_nfe_v2.py</code>	curvature statistics, correlations, stratified FID (Tables 2, 3; Figs 2–4)
<code>three.space.geometry.py</code>	arc/chord and correlations in $\mathcal{Z}$, $\mathcal{X}$, $\mathcal{Q}$
<code>gain_disentangle.py</code>	output-map gain variants and the amplification control
<code>curvature_schedule.py</code>	step-schedule comparison (Table 4)
<code>guidance_control.py</code>	teacher guidance sweep and reflow comparison (Table 5, Fig 5)
<code>reflow_experiment.py</code>	trains the reflow student
<code>make_figs_cluster.py</code>	all figures, from the saved JSON

2 The three spaces, as implemented

$\mathcal{Z}$: the state the ODE integrates. For LFM this is the standardized continuous encoder output, shape 49×256 . **For CDFM there is no separate $\mathcal{Z}$** : the sampler integrates 196×263 directly, so $\mathcal{Z}$ and $\mathcal{X}$ are the same tensor. This is why the two rows coincide exactly in the three-space table, and it is a definitional identity, not a measurement.

$\mathcal{X}$: the 196×263 motion representation. Obtained as `rvq.decoder(z * z_std + z_mean)` for LFM and as the state itself for CDFM. Note the unstandardization is part of the map.

$\mathcal{Q}$: Cartesian joints, $196 \times 22 \times 3$, obtained by `_gj` (line 390), which is `recover_from_ric(mn * std_t + mean_t)`. So $\mathcal{K}$ as implemented includes the un-normalization by the dataset statistics as well as the forward-kinematic recovery.

2.1 Norms

Space	Norm used	In code
$\mathcal{Z}$	Euclidean on the flattened state	<code>(a-b).flatten(1).norm(dim=1)</code>
$\mathcal{X}$	Euclidean on the flattened state	same
$\mathcal{Q}$	frame- and joint-averaged RMS, valid frames only	<code>d_q</code>

Explicitly, for $\mathcal{Q}$,

$$\|Q\|_{\text{rms}}^2 = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \frac{1}{J} \sum_j \|q_{\tau,j}\|_2^2,$$

with $\mathcal{T}$ the frames before padding. **Point for discussion:** $\mathcal{Z}$ and $\mathcal{X}$ use an unnormalized Euclidean norm while $\mathcal{Q}$ uses a frame-averaged one, so the three are not on a common scale. Arc-over-chord is a ratio and therefore dimensionless, so the comparison across spaces is still valid; but any statement about absolute distances is not. The turning angle is likewise scale-free. I do not think this affects any reported conclusion, but it should be stated in the setup.

3 Definitions of each statistic

Arc-over-chord $\mathcal{R}$. Summed step displacement divided by endpoint distance, in the norm of the space concerned, on the 50-step reference trajectory. Implemented as `arc_chord`. Equals 1 for a straight path.

Maximum turning angle. $\max_k \arccos(\langle \Delta_k, \Delta_{k+1} \rangle / (\|\Delta_k\| \|\Delta_{k+1}\|))$ over consecutive step displacements $\Delta_k = Y_{t_{k+1}} - Y_{t_k}$, cosine clamped to $\pm(1 - 10^{-6})$. Measured from the *extension* of the incoming segment, so a straight path gives 0. **Resolution-dependent:** it tends to 0 as $n \rightarrow \infty$ for any smooth path, so it is comparable only at a fixed grid; we always use $n = 50$.

Output-map gain. Finite-difference Jacobian-vector products at the terminal state with $P = 4$ random unit probes and $\varepsilon = 10^{-2}$: $\hat{g} = P^{-1} \sum_p \|\Psi(Y_1 + \varepsilon v_p) - \Psi(Y_1)\|_{\text{rms}} / \varepsilon$. **This estimates a mean, not a maximum, singular value**, because the probes are isotropic; it is nearer a normalized Frobenius norm than the operator norm in the bound. Variants at other evaluation points (trajectory mean, trajectory max, segment midpoint) are in `gain_disentangle.py`.

Velocity variation. Mean deviation of the velocity field along the trajectory from its trajectory average, normalized by that average — the straightness measure from the rectified-flow literature.

Degradation. Per clip, MPJPE between the sample at n steps and the same clip’s 50-step sample *at identical initial noise*, which is enforced by passing the same `seed` to both calls. It is a per-sample quantity precisely because FID is not.

FID and R-precision. `fid_calc` and `rprec` (lines 501, 505), on features from `memb` (line 499), the frozen motion evaluator. R@3 is `rprec(...)[3]`.

4 Seeding, clip selection, and what is averaged over what

- **Clip set.** Fixed across every experiment: `np.random.default_rng(0)`, first 512 of a permutation of the test indices, sorted. The same clips in the same order everywhere.
- **Sampling seed.** Within a batch starting at index s and replication r , the seed is $s + 100000r$. So a given clip receives the same noise at every step budget within a replication, which is what makes the degradation measure paired, and different noise across replications.
- **Two levels of averaging.** Curvature statistics and correlations are computed *per clip*, then a single number per replication is formed (a mean, or a Spearman coefficient across clips), and the confidence interval is taken across replications. So the intervals describe seed-to-seed variability, not clip-to-clip. The clip-to-clip spread is reported separately as the standard deviation, and it is the quantity of interest in the regularity result.

- **Replication counts.** Five seeds for curvature, correlations, schedules and the three-space study; three for the reflow and guidance comparisons. Intervals are t -based with $t_{0.975, R-1}$.

5 Discrepancies and open points

These are the items I would like to resolve with you.

1. The $\mathcal{Q}$ turning angle is probably wrong. In $\mathcal{Q}$ the joint tensor is flattened before the angle is computed, so the result is an angle between 66-dimensional displacement vectors and individual joints' turns partially cancel. This is the likely cause of the non-monotonicity ($0.358 \rightarrow 0.406 \rightarrow 0.347$ across the three spaces for LFM) while arc-over-chord rises monotonically. **Proposal:** either omit the $\mathcal{Q}$ turning angle or define it per joint and aggregate explicitly. I would not report the non-monotonicity as a finding.

2. $\mathcal{K}$ as implemented includes un-normalization. `_gj` multiplies by the dataset standard deviation and adds the mean before the kinematic recovery. That affine step is part of what we are calling the kinematic map. It is a diagonal rescaling, so it cannot bend a path, but it does change the metric, and the setup should say that $\mathcal{K}$ denotes the composition.

3. The gain is not the Lipschitz constant. As above, the probe estimates a mean singular value at a single point, whereas the bound's constant is a supremum over the region between the exact and approximate endpoints. The segment-evaluated variant matches the bound's structure for CDFM but not for LFM, and I have no explanation for the difference.

4. Degradation is reported in MPJPE, the analysis is stated in the RMS norm. They are equivalent up to a constant and order clips near-identically (rank correlation 0.993–0.997), so no conclusion changes, but the two sides of the inequality are currently measured in different norms.

5. The schedule experiment reads a stale profile. `curvature.schedule.py` loads the decoded arc-length profile from `curvature_nfe.json`, the output of the *first* curvature run, not `curvature_nfe.v2.json`. The profiles are close and the conclusion is a null result either way, but it should be regenerated for consistency before we report it.

6. Evaluation is on 512 clips, not the full set. Chosen for cost. The main replicated table elsewhere in the project uses 1024, and a full-set control at 2644 was run separately; FID is biased upward at smaller sample sizes, so absolute values are not comparable across those protocols and we should label them distinctly.

6 Availability

I am free for the walkthrough at your convenience; the scripts above are on the cluster under `$HOME/motion/code` and I can share the repository beforehand if that helps. Once we have gone through these points I will align the implementation with your generalized three-space setup and re-run anything whose definition changes.

With best regards,
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